Nexus Epigenomics – Investment Memo

Nexus is a bet that its enzyme-based epigenomic profiling platform targeting both DNA methylation (MeTLAND) and histone modifications (HisTLAND) can establish itself in the epigenomic sequencing market which is already moving away from the bisulfite sequencing pushed by better-funded competitors. Therefore, the field has already agreed the direction on which to move forward with the key question for Nexus to be whether they have a enough differentiated product to make it.

Core investment question: Can Nexus Epigenomics demonstrate, that their current productsm MeTLAND and HisTLAND offer an advantage both at the technical and cost level over the existing enzymatic alternatives that can be pushed either in to a commercial or licensable product before other better funded competitors like NEB further consolidate their market share?

1. Company Snapshot

Nexus Epigenomics is a Swedish startup that spun out of the Karolinska Institute, with two commercial products: HisTLAND, targeting histone modifications (commercially available), and MeTLAND, targeting DNA methylation (available on request). Both are positioned as gentler, enzyme-based alternatives to the bisulfite conversion methods that have dominated epigenomic profiling for two decades.

Funding to date comes from KI Innovations, SU Ventures, and Vinnova. The most significant grant is an ERC Proof of Concept award (€150K gross, €105K net contribution), intended to develop a novel enzyme-based method for detecting methylated DNA regions at lower cost and with better sample integrity than bisulfite. Vinnova is separately contributing €500K to validate MeTLAND for clinical samples, developing robust SOPs, a tailored data-analysis pipeline, and pilot-user engagement in translational and biopharma settings.

2. Core Problem

Epigenomic profiling is undergoing a technology transition. Bisulfite sequencing has been the standard method for studying DNA methylation for over twenty years, but its core limitations are well-documented and increasingly limiting: the chemical treatment degrades a large proportion of input DNA, making low material samples unreliable. Furthermore, it cannot distinguish between 5-methylcytosine (5mC) and 5-hydroxymethylcytosine (5hmC), two biologically distinct modifications with different roles in gene regulation.

Histone modifications are different to DNA methylation and thus, presents itself with a related but separate challenge. Profiling which histones are modified, and where across the genome, requires labour intensive methods and many cells that are not always possible to obtain. As epigenomics moves from a research tool toward clinical and translational applications, it needs to meet method standards that current approaches struggle to meet.

These are the two problems Nexus is targeting: MeTLAND addresses the DNA methylation detection problem, and HisTLAND addresses histone modification profiling. Both products are positioned on the same axis, gentler chemistry, better sample preservation, and compatibility with existing sequencing infrastructure, which is also the approach taken by other companies like NEB and Bimodal.

3. Technology & Scientific Differentiation

The founding team is a genuine scientific strength. CEO Mukund Kabbe is first author on a recent Nature Neuroscience paper in epigenomics, reflecting hands-on expertise in the field. The company also has the backing of Gonçalo Castelo-Branco and Marek Bartosovic, the co-inventor of MeTLAND. COO/CTO Jana Bartosovic brings direct industry scaling experience, having been instrumental in building operations at CARTANA AB and subsequently driving technology development and early-access programs at 10x Genomics following CARTANA's acquisition.

MeTLAND's core claim is that it can detect DNA methylation, including the 5mC/5hmC distinction that bisulfite cannot make using an enzyme-based approach that avoids the harsh DNA-degrading chemistry of bisulfite conversion. This is compatible with most existing sequencing platforms and is intended to work from low-input samples, making it potentially useful for clinical applications where material is scarce.

HisTLAND targets a different layer of epigenetic regulation, histone modification profiling and is already commercially available, giving Nexus a product in the market ahead of MeTLAND's full commercial launch.

The core information gap remains the same one that limits any early assessment: there is no public comparative data showing MeTLAND outperforming other current technologies such as bisulfite, EM-seq, or Biomodal's meaning that the independence confirmation of Nexus' technology working as advertised is still missing. This is the central open question for the investment case.

4. Market & Competitive Landscape

Industry structure

Epigenomics analysis is transforming itself from a pure research oriented area towards a more clinical, patient focused product relevant for widespread diseases such as cancer. While the shift from the well long-established bisulfite technology is already accepted, the sector is still missing the precision medicine require for this shift to happen in a economic viable manner. Therefore, a competitor will gain an advantage if is able to show validated clinical-sample workflows, compatible with existing lab infrastructure, that is economically viable and meets all the regulatory standards.

Competiors

The two most directly competitive products are biomodal's duet chemistry (combined genetic and epigenetic sequencing from a single sample) and NEB's EM-seq (enzymatic, non-bisulfite methylation detection). Both are essentially targeting the same space as MeTLAND. The additional context worth noting is that single-molecule long-read platforms from Oxford Nanopore and PacBio can detect methylation natively, without any conversion step, a longer-term structural challenge for the whole conversion-based category.

Competitor positioning

Company	Positioning	Advantage	Risk to Nexus
Illumina	Sequencing platform	Dominant established player with broad genomics and epigenetics portfolio	Default choice for most labs; sets the standard pipelines Nexus must be compatible with
Zymo Research	Established methylation kits and services	Deep customer trust, mature bisulfite and enzymatic workflows	Targets the same research and translational labs as Nexus
biomodal (former Cambridge Epigenetix)	Combined genetic and epigenetic sequencing from a single low-input sample	$88M raised in Series D; strong liquid-biopsy positioning, avoids harsh chemical treatment	Best-funded direct conceptual competitor; pursuing the same gentle-chemistry value proposition
New England Biolabs (EM-seq)	Enzymatic non-bisulfite methylation detection	Deep reagent distribution, trusted brand, similar mild-chemistry narrative	Practically identical technical pitch to Nexus, backed by a much larger company
Oxford Nanopore / PacBio	Single-molecule native methylation detection	No conversion step at all; reads genetic and epigenetic signals simultaneously	Longer-term platform threat that could make conversion-based methods less relevant
Diagenode (Hologic)	End-to-end epigenomics services	Diagnostics-company scale and distribution across ChIP-seq, ATAC-seq, and methylation	Competes for core-lab and CRO services revenue Nexus would need to capture early on

Nexus Epigenomics positioning

Strengths

- Founding team with deep, hands-on epigenomics expertise from Karolinska Institute, directly relevant to both products
- MeTLAND's enzyme-based approach addresses real, well-documented limitations of bisulfite sequencing
- HisTLAND already commercially available, providing early market presence
- Vinnova grant specifically targeting clinical-sample validation — the most commercially relevant proof point
- Jana Bartosovic's operational track record at CARTANA and 10x Genomics reduces execution risk at early stage

Weaknesses

- Significantly earlier than direct competitors: biomodal is Series D-funded, NEB has established distribution at scale

- The gentle-chemistry, low-input methylation pitch is not unique — NEB and biomodal are making essentially the same claim with more resources behind it
- No public head-to-head data showing MeTLAND outperforming EM-seq or biomodal on any concrete metric
- No disclosed commercial customers beyond the Vinnova-funded pilot framework
- Two-person team limits bandwidth for simultaneous product development, commercial outreach, and regulatory groundwork

Structural insight: The core question for Nexus is not whether milder, enzyme-based methylation profiling is better than bisulfite, the sector has already agreed on that and moving towards alternatives. What is currently missing is to whether MeTLAND has a genuine technical or cost edge over those two better-resourced alternatives, and whether Nexus can get from grant-funded lab tool to validated clinical workflow before one of the competitors gets established as the "go-to technology" similar to for example PacBio and long-read sequencing.

How the market could look in 5–7 years

Scenario A: Consolidation around 2–3 comapanies (Most Likely)

EM-seq, biomodal's duet chemistry, or native long-read detection becomes the default, and smaller tool developers either get acquired or license into one of these dominant platforms. Nexus's path in this scenario runs through demonstrating a concrete superiority on cost-per-sample or clinical-sample performance, not novelty alone. Early clinical validation through the Vinnova project would be the most credible entry point. It can also survive as a local start up within the university campues environment that provides full on support to surrounding companies/ labs.

Scenario B: Native long-read sequencing bypasses conversion chemistry entirely

If Nanopore or PacBio-style native detection becomes cheap and standard enough for routine use, the whole current technology proposed by Nexus, NEB, and biomodal will get reduced to a niche. However, the timeline for this is uncertain but the directional risk is real, particularly in research settings where instrument cost is less of a constraint.

Scenario C: Services and licensing rather than independent platform scale

Given Nexus's size and resource constraints, a plausible and not-unfavorable outcome is MeTLAND being licensed to or acquired by a larger reagent or diagnostics company, similar to CARTANA's acquisition by 10x Genomics. In this scenario, Nexus doesn't need to build independent commercial scale; it needs to generate enough validated differentiation to be worth acquiring.

Scenario C is probably the most realistic near-term path to value realization for Nexus, and the team's track record makes it a credible one. Scenarios A and B both require either outcompeting better-funded players or surviving a platform transition, which is harder to underwrite at this stage.

5. Moat Analysis

Nexus's defensibility is best understood as a conditional stack, but the layers look different for an early-stage tools company than for a therapeutics company.

Patents (weakest layer)

Enzymatic and mild-chemistry methylation detection is already a crowded place when it comes to intellectual property. For example, a quick search comes with the fact that NEB holds patents around EM-seq, biomodal holds patents around its duet and modC chemistry, and the broader category of avoiding bisulfite has substantial prior art across academic and commercial groups. Minor protocol variations can, furthermore, circumvent method patents meaning that Nexus potentially does not have the current IP power to fill the depth or litigation required to defend their IP against bigger competitors.

Technical and process know-how (core layer)

The more durable near-term advantage, if it exists, would come from MeTLAND's actual assay performance: DNA integrity preservation relative to bisulfite and EM-seq, reliable low-input performance, batch-to-batch reproducibility, and high compatibility with existing sequencing technologies and bioinformatics pipelines There is no current public data regarding that combability.

Customer lock-in and regulatory validation

In life-science, the strongest advantage is the fact that if a customer standardizes its workflow, analysis pipeline, and validated SOPs around one methylation method, switching is disruptive even if a competitor is marginally better. Nexus has none of this yet. If the Vinnova-backed work succeeds in producing validated clinical-sample SOPs, a second, stronger advantage could open. But this is conditional on clinical validation happening first.

6. How the thesis could break down

Layer	Primary Failure Risk
Technical	MeTLAND fails to show a significance better performance over EM-seq or biomodal or shows reproducibility issues at low input. DNA-integrity preservation claims do not hold up under independent testing
Commercial / Adoption	Research labs and core facilities are already standardized on Illumina, Zymo, or NEB workflows and have no real reason to switch for an unproven early-stage startup. Nexus lacks the sales infrastructure to generate that pull
Competitive / Platform	Native long-read sequencing increasingly detects methylation without any conversion chemistry, shrinking the addressable category for conversion-based methods broadly where Nexus market is present.
Clinical / Regulatory	Vinnova-backed clinical-sample validation fails to produce a reproducible, regulatory-viable SOP; precision-medicine pilot partners do not materialize or do not renew
Capital	The company cannot raise a substatial round before larger competitors further commoditize the gentle enzymatic methylation category

7. Key Catalysts

Near-term

- Peer-reviewed or preprint publication of MeTLAND performance data, ideally head-to-head against bisulfite and EM-seq

- Completion of Vinnova project deliverables: validated clinical-sample SOPs and a tailored analysis pipeline
- A priced seed round beyond current incubator-level backing
- Patent filings specific to MeTLAND's mechanism, to the extent it is genuinely novel

Medium-term

- Demonstrated cost-per-sample or input-sensitivity advantage over NEB and biomodal in independent hands
- Academic medical center or CRO partnership generating real-world validation data
- Expansion beyond DNA methylation into chromatin accessibility or broader histone profiling use cases
- Evidence that MeTLAND performs in single-cell or low-biomass clinical applications specifically, where competitors are comparatively weaker

8. Investment Thesis

Nexus Epigenomics is a very early, high-risk bet that a small but scientifically credible team can build a differentiated enzyme-based epigenomic profiling platform at a moment when the field is already moving away from bisulfite conversion. However, it currently stands several years and at least one funding round behind the competitors already making the same transition.

The core hypothesis is not that gentler enzymatic methylation detection is a good idea, the market has already validated that. It is that MeTLAND specifically offers a meaningful edge in cost, input sensitivity, or clinical-sample compatibility sufficient to win share or licensing interest from much better-resourced players pursuing the same basic thesis.

This sits earlier and higher variance than most comparable bets, closer to a pre-seed technical wager than a company with a near-term commercial or clinical pathway. A rough estimate would put the probability of Nexus reaching a validated, differentiated, commercial or licensable product within ten years, with the more likely favorable outcome being a technology license or acquisition by a larger reagent or diagnostics company rather than independent platform scale. The team's CARTANA and 10x Genomics track record makes that path credible.

9. When to change your mind

The investment opportunity Nexus offers will weaken if after 18-24 months, various of the following conditions are met:

- No independent, peer-reviewed, or third-party comparative data showing MeTLAND outperforming bisulfite, EM-seq, or biomodal
- No named partnerships, or academic collaborators beyond the Vinnova-funded project
- No funding round, with the company still reliant solely on incubator and grant-level support
- Vinnova project fails to produce a validated clinical-sample SOP, or produces one with no clear path to real-world adoption
- No clearer articulation of technical differentiation versus NEB EM-seq or biomodal's duet chemistry beyond marketing claims

If most of these persist beyond the evaluation window, the probability-weighted case drops sharply, this sits closer to monitor, not invest than to a company with identifiable near-term de-risking events.

10. Bull Case

- Independent data confirms MeTLAND offers a real advantage in low-input samples such as liquid biopsy or degraded clinical samples
- Vinnova-backed clinical validation opens a precision-medicine or diagnostic pathway ahead of larger, slower-moving competitors
- They secure seed round plus at least an academic medical center
- Nexus becomes an acquisition target for a larger company that wants to add their technology to their portfolio
- HisTLAND is established as a real commercial alternative which enables MeTLAND to get further developed without increasing capital dependency
- Platform expands credibly into chromatin accessibility or broader histone profiling, widening the addressable use cases beyond DNA methylation

11. Bear Case

- MeTLAND shows no measurable advantage over EM-seq or biomodal's chemistry once tested independently
- Potential clients see no compelling reason to switch from already-standardized workflows for an unproven early-stage product
- Native long-read sequencing makes conversion-based methylation detection broadly less relevant before Nexus can establish a foothold
- Company fails to raise a priced round and remains capacity-constrained, unable to generate the validation data needed to compete
- Vinnova-funded clinical validation stalls or produces SOPs that do not achieve real-world adoption
- Larger competitors further commoditize the gentle enzymatic methylation positioning before Nexus can differentiate on a concrete technical axis

12. Conclusion

The central question for Nexus Epigenomics is whether a very early Swedish startup can establish a defensible position in DNA methylation and histone profiling at a moment when the category's core value proposition, gentler, non-bisulfite chemistry has already been validated and commercially captured by much larger, better-funded players.

The scientific premise is sound, and the founding team is credible. What is unresolved is whether MeTLAND has any independently verifiable technical edge over NEB's EM-seq or biomodal's duet chemistry, whether the Vinnova-backed clinical validation work can produce genuine traction, and whether the company can do any of this before incumbents further consolidate the category or long-read platforms reduce its relevance.